

02_Partial Derivative

Rules in Partial Derivatives

- 1.Constant Rule
- 2.Power Rule
- 3.Sum and Difference Rule
- 4.Product Rule
- 5.Quotient Rule
- 6.Chain Rule
- 7.Mixed Partial Derivative Rule (Clairaut's Theorem)
- 8.Higher-Order Derivatives Rule
- 9.Implicit Differentiation Rule

1. Power Rule

For a function $f(x, y) = x^n$ or $f(x, y) = y^n$:

$$\frac{\partial}{\partial x}(x^n) = nx^{n-1}$$

$$\frac{\partial}{\partial y}(y^n) = ny^{n-1}$$

Example:

$$f(x, y) = x^3$$

$$\frac{\partial f}{\partial x} = 3x^2$$

$$g(x, y) = y^4$$

$$\frac{\partial g}{\partial y} = 4y^3$$

2. Constant Multiple Rule

For a function $f(x, y) = c \cdot g(x, y)$, where c is a constant:

$$\frac{\partial}{\partial x}(c \cdot g(x, y)) = c \cdot \frac{\partial g(x, y)}{\partial x}$$

$$\frac{\partial}{\partial y}(c \cdot g(x, y)) = c \cdot \frac{\partial g(x, y)}{\partial y}$$

Example:

$$f(x, y) = 5x^2$$

$$\frac{\partial f}{\partial x} = 5 \cdot 2x = 10x$$

$$g(x, y) = 7y^3$$

$$\frac{\partial g}{\partial y} = 7 \cdot 3y^2 = 21y^2$$

3. Sum Rule

For a function $f(x, y) = g(x, y) + h(x, y)$:

$$\frac{\partial}{\partial x}(g(x, y) + h(x, y)) = \frac{\partial g(x, y)}{\partial x} + \frac{\partial h(x, y)}{\partial x}$$
$$\frac{\partial}{\partial y}(g(x, y) + h(x, y)) = \frac{\partial g(x, y)}{\partial y} + \frac{\partial h(x, y)}{\partial y}$$

Example:

$$f(x, y) = x^2 + y^2$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial x}(y^2) = 2x + 0 = 2x$$

$$g(x, y) = 3x^3 + 4y$$

$$\frac{\partial g}{\partial y} = \frac{\partial}{\partial y}(3x^3) + \frac{\partial}{\partial y}(4y) = 0 + 4 = 4$$

4. Product Rule

For a function $f(x, y) = g(x, y) \cdot h(x, y)$:

$$\frac{\partial}{\partial x}(g(x, y) \cdot h(x, y)) = g(x, y) \cdot \frac{\partial h(x, y)}{\partial x} + h(x, y) \cdot \frac{\partial g(x, y)}{\partial x}$$

$$\frac{\partial}{\partial y}(g(x, y) \cdot h(x, y)) = g(x, y) \cdot \frac{\partial h(x, y)}{\partial y} + h(x, y) \cdot \frac{\partial g(x, y)}{\partial y}$$

Simplified,

$$\frac{\partial f}{\partial x} = u \cdot \frac{\partial v}{\partial x} + v \cdot \frac{\partial u}{\partial x}$$

$$\frac{\partial f}{\partial x} = u \cdot \left(\frac{\partial}{\partial x} v \right) + v \cdot \left(\frac{\partial}{\partial x} u \right)$$

Example:

$$f(x, y) = (x^2) \cdot (y^3)$$

$$\frac{\partial f}{\partial x} = (x^2) \cdot \frac{\partial (y^3)}{\partial x} + (y^3) \cdot \frac{\partial (x^2)}{\partial x} = (x^2) \cdot 0 + (y^3) \cdot (2x) = 2xy^3$$

$$g(x, y) = (3x) \cdot (4y^2)$$

$$\frac{\partial g}{\partial y} = (3x) \cdot \frac{\partial (4y^2)}{\partial y} + (4y^2) \cdot \frac{\partial (3x)}{\partial y} = (3x) \cdot (8y) + (4y^2) \cdot 0 = 24xy$$

Partial Derivative with Respect to x , $f(x, y) = (x^2 + y^2)(x^2y + y^2x)$

First, we compute the partial derivatives of $u(x, y) = x^2 + y^2$ and $v(x, y) = x^2y + y^2x$ with respect to x :

$$u(x, y) = x^2 + y^2$$

$$v(x, y) = x^2y + y^2x$$

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x}(x^2y + y^2x) = 2xy + y^2$$

$$\frac{\partial f}{\partial x} = u \cdot \frac{\partial v}{\partial x} + v \cdot \frac{\partial u}{\partial x}$$

$$\frac{\partial f}{\partial x} = (x^2 + y^2)(2xy + y^2) + (x^2y + y^2x)(2x)$$

Final Answer: $4x^3y + 3x^2y^2 + 2xy^3 + y^4$

Another approach:

$$f(x, y) = x^2(x^2y + y^2x) + y^2(x^2y + y^2x)$$

$$f(x, y) = x^4y + x^3y^2 + x^2y^3 + xy^4$$

$$\frac{\partial f}{\partial x} = 4x^3y + 3x^2y^2 + 2xy^3 + y^4$$

Partial Derivative with Respect to y . $f(x, y) = (x^2 + y^2)(x^2y + y^2x)$

Now, we compute the partial derivatives of $u(x, y)$ and $v(x, y)$ with respect to y :

$$\frac{\partial u}{\partial y} = 2y$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y}(x^2y + y^2x) = x^2 + 2yx$$

Applying the product rule:

$$\frac{\partial f}{\partial y} = u \cdot \frac{\partial v}{\partial y} + v \cdot \frac{\partial u}{\partial y}$$

$$\frac{\partial f}{\partial y} = (x^2 + y^2)(x^2 + 2yx) + (x^2y + y^2x)(2y)$$

Final Answer: $\frac{\partial f}{\partial y} = x^4 + 2x^3y + 3x^2y^2 + 4xy^3$

Partial Derivative with Respect to y . $f(x, y) = (x^2 + y^2)(x^2y + y^2x)$

Now, we compute the partial derivatives of $u(x, y)$ and $v(x, y)$ with respect to y :

Multiply both expressions:

$$f(x, y) = (x^2)(x^2y) + (x^2)(y^2x) + (y^2)(x^2y) + (y^2)(y^2x)$$

$$\frac{\partial f}{\partial y} = x^4 + 2x^3y + 3x^2y^2 + 4xy^3$$

(Final Answer)

5. Quotient Rule

For a function $f(x, y) = \frac{g(x, y)}{h(x, y)}$:

$$\frac{\partial}{\partial x} \left(\frac{g(x, y)}{h(x, y)} \right) = \frac{h(x, y) \cdot \frac{\partial g(x, y)}{\partial x} - g(x, y) \cdot \frac{\partial h(x, y)}{\partial x}}{[h(x, y)]^2}$$

$$\frac{\partial}{\partial y} \left(\frac{g(x, y)}{h(x, y)} \right) = \frac{h(x, y) \cdot \frac{\partial g(x, y)}{\partial y} - g(x, y) \cdot \frac{\partial h(x, y)}{\partial y}}{[h(x, y)]^2}$$

Example:

$$f(x, y) = \frac{x^2}{y}$$

$$\frac{\partial f}{\partial x} = \frac{y \cdot \frac{\partial (x^2)}{\partial x} - x^2 \cdot \frac{\partial y}{\partial x}}{y^2} = \frac{y \cdot (2x) - x^2 \cdot 0}{y^2} = \frac{2xy}{y^2} = \frac{2x}{y}$$

$$g(x, y) = \frac{y^2}{x}$$

$$\frac{\partial g}{\partial y} = \frac{x \cdot \frac{\partial (y^2)}{\partial y} - y^2 \cdot \frac{\partial x}{\partial y}}{x^2} = \frac{x \cdot (2y) - y^2 \cdot 0}{x^2} = \frac{2xy}{x^2} = \frac{2y}{x}$$

Simplified,

$$f(x, y) = \frac{u(x, y)}{v(x, y)}$$

$$\frac{\partial f}{\partial x} = \frac{v \cdot \frac{\partial u}{\partial x} - u \cdot \frac{\partial v}{\partial x}}{v^2}$$

$$\frac{\partial f}{\partial x} = \frac{v \cdot \left(\frac{\partial}{\partial x} u \right) - u \cdot \left(\frac{\partial}{\partial x} v \right)}{v^2}$$

6. Chain Rule

If, $f(x) = h(g(x)) \rightarrow$ Then, $f'(x) = h'(g(x)) \cdot g'(x)$

$$f(x) = (3x + 2)^4 \rightarrow$$

- Outer: $h(u) = u^4$
- Inner: $u = g(x) = 3x + 2$

$$\frac{df}{dx} = h'(g(x)) \cdot g'(x) = 4(3x + 2)^3 \cdot 3 = 12(3x + 2)^3$$

Example:

$$f(x, y) = e^{x^2+y^2}$$

Let $u = x^2 + y^2$, then $f = e^u$.

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial f}{\partial u} = e^u$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} = e^{x^2+y^2} \cdot 2x = 2xe^{x^2+y^2}$$

$$= e^{x^2+y^2} \cdot \frac{d}{dx}(x^2 + y^2) = e^{x^2+y^2} \cdot (2x)$$

$$\boxed{\frac{\partial f}{\partial x} = 2x \cdot e^{x^2+y^2}}$$

Example: $\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i - y_{\text{pred}})^2$ **Here,** $y_{\text{pred}} = mx_i + c$

Plug into the cost function: $\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i - (mx_i + c))^2$

Take the derivative of Cost w.r.t. m : $\frac{\partial \text{Cost}}{\partial m} = \frac{\partial}{\partial m} \left[\frac{1}{n} \sum_{i=0}^n (y_i - (mx_i + c))^2 \right]$

$$= \frac{1}{n} \sum_{i=0}^n 2(y_i - (mx_i + c))(-x_i)$$

$$= \frac{-2}{n} \sum_{i=0}^n x_i (y_i - y_{\text{pred}})$$

Final Answer: $\frac{\partial \text{Cost}}{\partial m} = -\frac{2}{n} \sum_{i=0}^n x_i (y_i - y_{\text{pred}})$

Example: $\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i - y_{\text{pred}})^2$ **Here,** $y_{\text{pred}} = mx_i + c$

Plug into the cost function: $\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i - (mx_i + c))^2$

$$= y_i^2 - 2y_i(mx_i + c) + (mx_i + c)^2$$

$$\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i^2 - 2y_i mx_i - 2y_i c + m^2 x_i^2 + 2mx_i c + c^2)$$

Take the derivative of Cost w.r.t. m : $\frac{d(\text{Cost})}{dm} = \frac{1}{n} \sum_{i=0}^n (-2y_i x_i + 2mx_i^2 + 2cx_i)$

$$= \frac{2}{n} \sum_{i=0}^n x_i (y_{\text{pred}} - y_i) \quad \left| \quad y_{\text{pred}} = mx_i + c \right.$$

Final Answer: $= -\frac{2}{n} \sum_{i=0}^n x_i (y_i - y_{\text{pred}})$

7. Mixed Partial Rule (Clairaut's Theorem)

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

Example:

$$f(x, y) = x^2 y^3 + xy$$

$$\frac{\partial f}{\partial x} = 2xy^3 + y$$

$$\frac{\partial^2 f}{\partial y \partial x} = 6xy^2 + 1$$

$$\frac{\partial f}{\partial y} = 3x^2 y^2 + x$$

$$\frac{\partial^2 f}{\partial x \partial y} = 6xy^2 + 1$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 6xy^2 + 1 \quad \text{(Final Answer)}$$

Some Important Calculus Formula:

Derivative	Integral (Antiderivative)	$\frac{d}{dx} \tan x = \sec^2 x$	$\int \sec^2 x \, dx = \tan x + C$
$\frac{d}{dx} n = 0$	$\int 0 \, dx = C$	$\frac{d}{dx} \cot x = -\csc^2 x$	$\int \csc^2 x \, dx = -\cot x + C$
$\frac{d}{dx} x = 1$	$\int 1 \, dx = x + C$	$\frac{d}{dx} \sec x = \sec x \tan x$	$\int \tan x \sec x \, dx = \sec x + C$
$\frac{d}{dx} x^n = nx^{n-1}$	$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$	$\frac{d}{dx} \csc x = -\csc x \cot x$	$\int \cot x \csc x \, dx = -\csc x + C$
$\frac{d}{dx} e^x = e^x$	$\int e^x \, dx = e^x + C$	$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$
$\frac{d}{dx} \ln x = \frac{1}{x}$	$\int \frac{1}{x} \, dx = \ln x + C$	$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$	$\int -\frac{1}{\sqrt{1-x^2}} \, dx = \arccos x + C$
$\frac{d}{dx} n^x = n^x \ln n$	$\int n^x \, dx = \frac{n^x}{\ln n} + C$	$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$	$\int \frac{1}{1+x^2} \, dx = \arctan x + C$
$\frac{d}{dx} \sin x = \cos x$	$\int \cos x \, dx = \sin x + C$	$\frac{d}{dx} \operatorname{arccot} x = -\frac{1}{1+x^2}$	$\int -\frac{1}{1+x^2} \, dx = \operatorname{arccot} x + C$
$\frac{d}{dx} \cos x = -\sin x$	$\int \sin x \, dx = -\cos x + C$	$\frac{d}{dx} \operatorname{arcsec} x = \frac{1}{x\sqrt{x^2-1}}$	$\int \frac{1}{x\sqrt{x^2-1}} \, dx = \operatorname{arcsec} x + C$
		$\frac{d}{dx} \operatorname{arccsc} x = -\frac{1}{x\sqrt{x^2-1}}$	$\int -\frac{1}{x\sqrt{x^2-1}} \, dx = \operatorname{arccsc} x + C$